

01_FINAL_VALIDATION_RESULTS

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EVIDENCE PROVENANCE: [SOFTWARE VERIFIED]

Automated execution of 3,160 tests across backend, frontend, and hardware contract test suites.

1. Master Automated Test Suite Summary

Subsystem Suite	Framework	Total Tests	Pass	Fail	Execution Time
Backend & Safety Solvers	pytest	1,213	1,213	0	18.42 s
Frontend & Cockpit HMI	vitest	1,860	1,860	0	12.15 s
Fog Governor Suite	pytest	21	21	0	1.01 s
Hardware Contracts	pytest	66	66	0	10.35 s
FULL WORKSPACE SUITE	Unified	3,160	3,160	0	100% PASS

2. Invariant Compliance Verification

- **Rule 3 (Hardware Honesty):** Zero fake hardware claims. Injected and derived data strictly classified.
- **Rule 5 (Single State Store):** Authoritative Digital Twin in twin_state_store.py is the sole source of truth.
- **Rule 6 (Frontend Purity):** Frontend consoles are pure presentation projections with zero physics calculation.
- **Rule 7 (Safety Invariant):** Local vehicle governor unconditionally clamps dispatch requests (v_command <= v_safe).